



Pontrilas 2026

A report on behalf of the Wye Salmon Association.

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A short note on the impact of Pontrilas STW effluent on the Dulas Brook, looking at 2025-26 data.

Map

Pontrilas STW discharges into the Dulas Brook at Pontrilas, about 200m above its Dore confluence.

Figure 1 shows a map of the area concerned. WSA have two relevant WQ monitoring sites, one upstream at Ewyas Harold, and a second at Pontrilas, about 150m below the STW discharge. Estimated annual mean flowrates are 2.1 m³/s for the Dore and 0.25 m³/s for the Dulas.



The STW at Pontrilas has an input capacity of 1640 PE and current load is 877 PE. Any impacts on the Dulas could double in future without any changes being required to the current discharge permit (AH1003301) which appears to be based on the capacity and not the current load.

The data

Data for this report has been downloaded from EpiCollect and covers the period Jan-2025 to Aug-2026.

For sewage, it is worth examining phosphate, ammonia and SC (specific conductivity, derived from TDS).

Specific conductivity

Sewage effluent contains a considerable quantity of dissolved material which will increase downstream conductivity of the river. This can be very significant for small streams, particularly in low summer flows. Figures 2a & 2b show the trend plots for SC through 2025 and up to early August of 2026 respectively. The upstream site in blue (W071 at Ewyas Harold) shows the natural behaviour of the Dulas, with low-flow SC around 500 $\mu\text{S}/\text{cm}$. Natural SC correlates inversely with flowrate, so in figure 2a we see a gentle rise as the river height drops through the spring and summer, as well as significant falls when the river rises in autumn & winter. The same pattern is seen in figure 2b.

The downstream site in green (WSA078 at Pontrilas) is noticeably higher and the divergence from WSA071 increases as the flow falls during the summer. This divergence is even more significant in the lower flowrates seen in 2026 (figure 2b). Effluent dilution is reduced in proportion to the flowrate. As of early August-26, effluent from Pontrilas STW is increasing SC from its natural level of 500 $\mu\text{S}/\text{cm}$ up to 770 $\mu\text{S}/\text{cm}$ – a 54% increase.

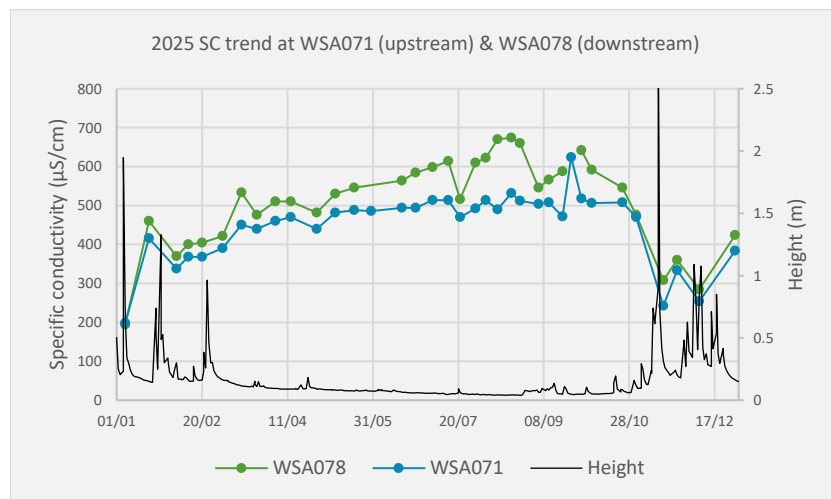
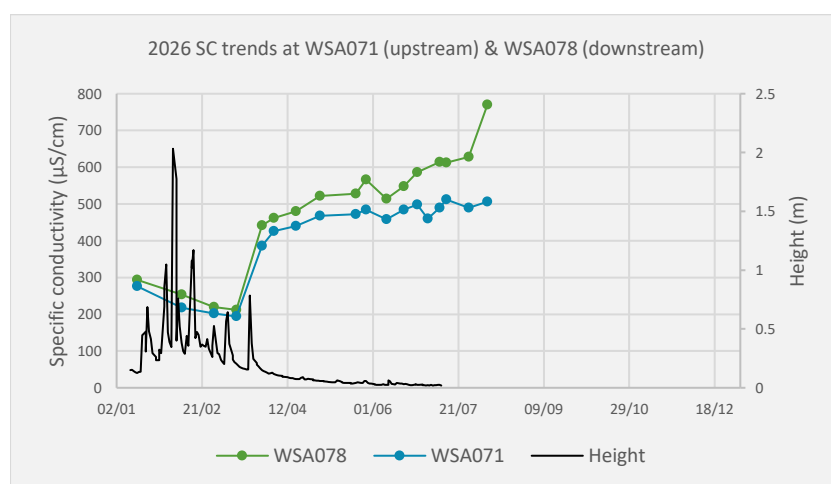


Figure 2:
(a) 2025 specific conductivity trend plot, showing data from WSA071 (upstream at Ewyas Harold) and WSA078 downstream of the Pontrilas STW.
In black we show the height of the Dulas Brook measured by EA at Ewyas Harold.



(b) as above for 2026.

If we assume that effluent has an SC of 1200 $\mu\text{S}/\text{cm}$ (a typical UK value I found on the internet), we can estimate that the Dulas at Pontrilas is 22% sewage effluent.

We can cross-check this number another way. Taking the current input load at Pontrilas of 877 PE and multiplying by 180 l/day (the typical liquid output per PE) gives an average flowrate of circa 2 l/s. Current flowrates (mid-Aug-26) on rivers in the area which have volumetric flow measurement are well below Q_{95} ; often around 5% of annual mean flowrate. Applying this to the Dulas, we can estimate current flowrate at circa 10 l/s, around 5x the STW output; this being totally consistent with our above dilution estimate.

Essentially, DCWW are using the Dulas as a convenient 'duct' between their STW and the much larger River Dore (circa 10x the flow of the Dulas). With climate change promising ever drier summers and the capacity to double the load at Pontrilas, this location has the potential to get extremely ugly.

Phosphate

Mean RP at Ewyas Harold (WSA071) was 0.062 mgP/l in 2025 and, so far, 0.062 mgP/l in 2026; numbers which would scrape into WFD good status (<0.07 mgP/l). Most of the RP load is diffuse agricultural, although there is a low-flow upswing presumably due to upstream septic tanks, etc. which raises the annual mean above the diffuse background. Figure 3a shows the 2025 trend plot at this site. In figure 3b the same data is plotted against river height to more clearly show the flow dependency.

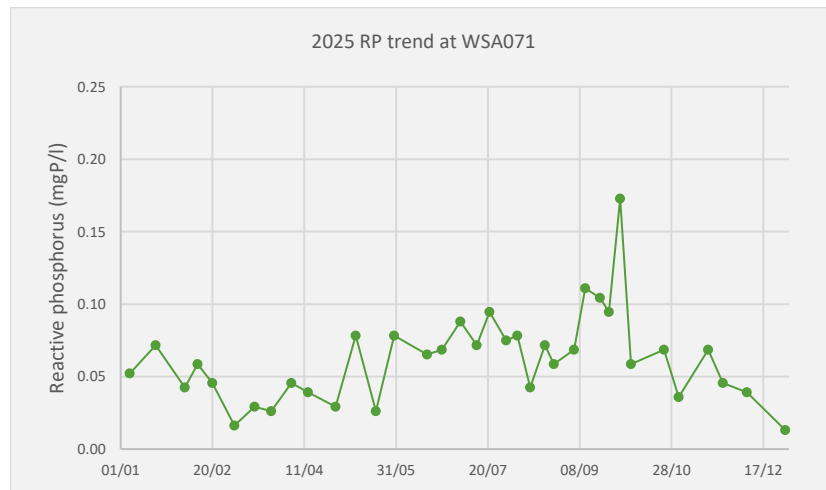
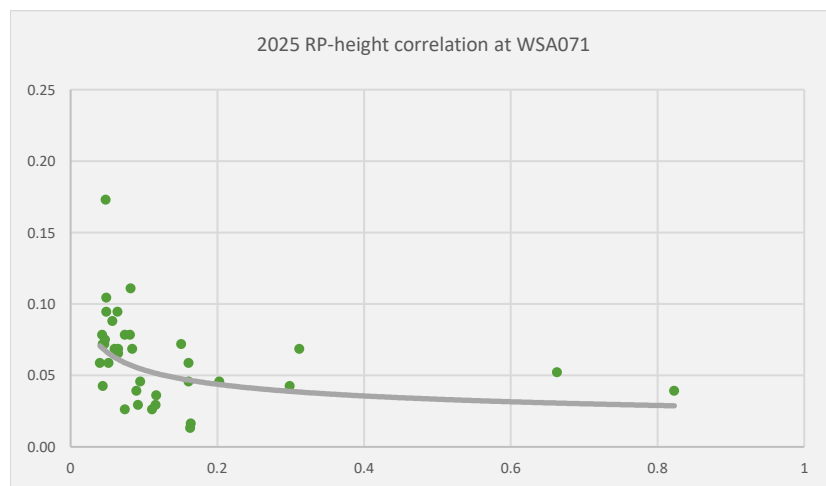


Figure 3:
(a) 2025 RP trend plot for WSA071
Ewyas Harold.



(b) the same data plotted against
river height; the grey curve is an
illustrative regression curve (power
law).

Downstream at Pontrilas (WSA078), the annual mean was 0.26 mgP/l in 2025 and is currently at 0.29 mgP/l in 2026; so WFD 'poor' status at this location.

Figure 4 shows the 2025 dataset plotted against river height. The chart is dominated by the dilution curve due to the STW load.

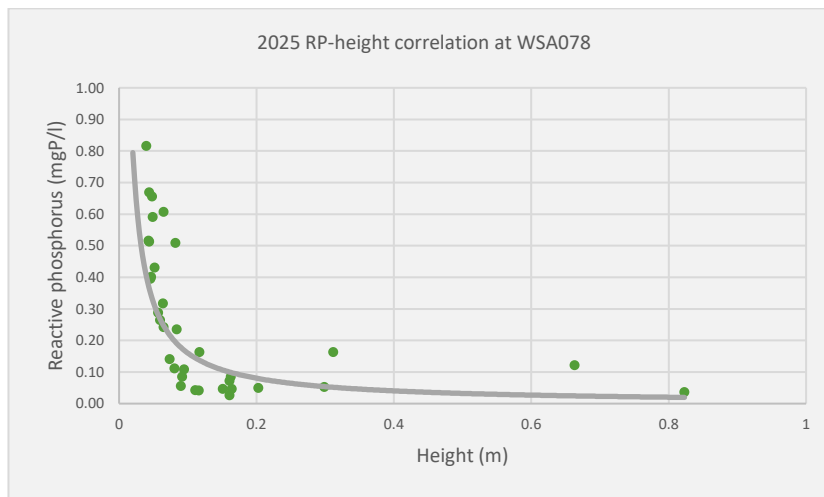


Figure 4: 2025 RP data from WSA078 plotted against river height; the grey curve is an illustrative regression curve (power law).

The 2026 situation is shown by the trend plot in figure 5. The massive divergence between the two sites is seen to increase progressively as flowrates fall through the summer. At August-26 flowrates, about 95% of the RP signal is coming from the STW.

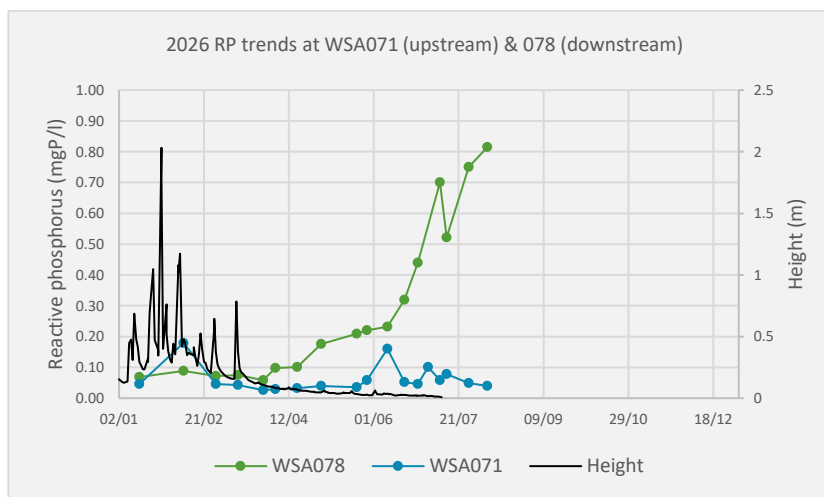


Figure 5: 2026 RP trend plot, showing data from WSA071 (upstream at Ewyas Harold) and WSA078 downstream of the Pontrilas STW. In black we show the height of the Dulas Brook measured by EA at Ewyas Harold.

Ammoniacal nitrogen

Ammoniacal nitrogen signals are similar at both sites. Nor is there any clear seasonality or flowrate dependence on this parameter. The 90%ile values for 2025 were 0.09 mgN/l at WSA071 and 0.13 mgN/l at WSA078. Both sites are comfortably compliant with WFD 'good' status (<0.6 mgN/l).

Conclusion

RP results for 2025 and 2026 are very similar to data from 2022-3 used in our earlier report on this location. The long-promised P-stripping for Pontrilas STW clearly is not yet operational.

The SC results, specifically the dilution estimates, highlight a wider issue. The problems at Pontrilas are not all about RP; we must consider all of the effluent content with respect to future dilution rates. Climate change is expected to increase pressure on base flows over coming years, plus Pontrilas has the capacity to output twice the effluent it currently does. This is a recipe for ecological devastation which is being played out at lots of rural STWs, not just at Pontrilas (Much Dewchurch and Peterchurch come immediately to mind). But at this location the whole problem could be averted by 200m of pipe to divert the effluent directly down to the much larger River Dore.